

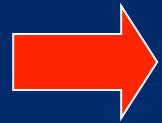
EE 558

How to Give a Technical Presentation:
Think Outside Your Little World

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Outline



1. Fundamental Value of a Good Presentation

2. Understand Your Audience

3. Basics of a Presentation

4. Good Practices

5. Questions and Answers

6. Always Try to Get Better

The Fundamental Value

- ✓ You will be “graded” on your presentations throughout your career.
- ✓ All students in my classes give 10-minute presentations.
- ✓ A poor presentation can kill a wonderful project, and vice-versa (unfortunately).

Value Your Presentation Time

- ✓ If you repeat a main idea a few times, then *maybe* the audience will remember it.
 - Tell them what you will tell them
 - Tell them
 - Tell them what you told them
- ✓ You have a 2-minute grace period of attention.
 - *meow, meow, meow*
 - *blah, blah, blah,*
 - *lunch, email, phone calls, etc.*
- ✓ Rehearse well.
 - *(10 minutes X 10) << (time on the project)*
 - *Ask for advice on slides from experts and novices.*

Respect Your Audience

- *Never* talk down to your audience.
- They are very bright, they just don't know as much as you do about your project.
- Right attitude: "I have some interesting ideas that I want to share."

Audience Motivation

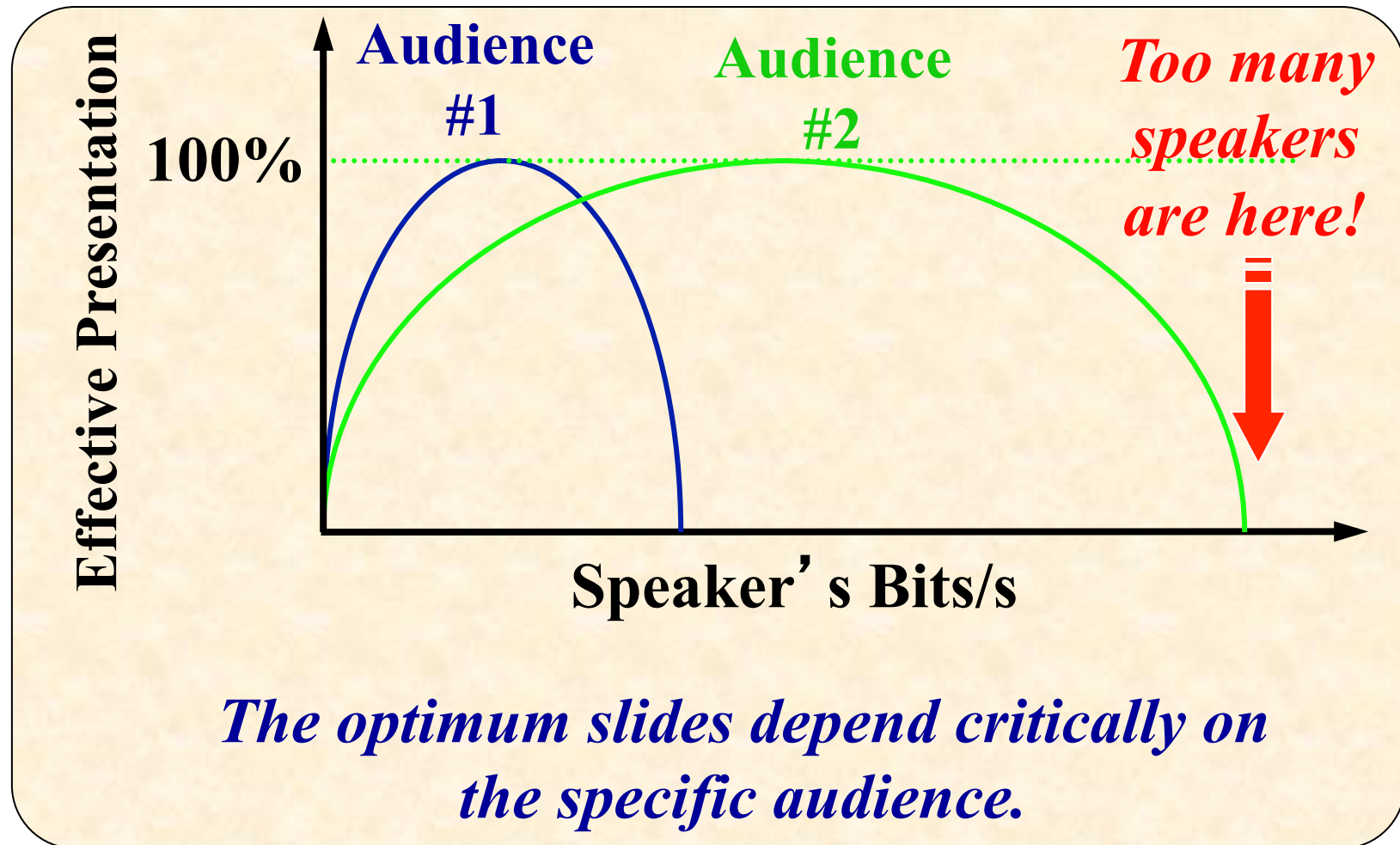
- Who's your customer?
 - Who's your customer's customer?
 - What does your audience want to learn?
- ❖ *I ask students in short courses what they want.*

Understand the Level of Audience

KISS: Keep it simple, stupid.

- ✓ Never assume a background.
- ✓ Once you lose your listener, they usually don't come back.
- ✓ Don't underestimate how much people like hearing what they already know.
- ✓ Practice with a non-expert friend, and then ask him/her questions afterwards.
 - You'll be surprised.

The Right Message for the Right Audience



- Every audience is different.
- Therefore, the presentation must be different.

Key Metrics

Five key figures-of-merits:

1. Content

- *Does your data support your claims?*

2. Coherence

- *Can the audience follow the thought train?*

3. Energy

- *If you are not excited about your topic, why should the audience?*

4. Timeliness

- *You love your topic, but does the audience?*
- *Don't fall in love with your slides.*

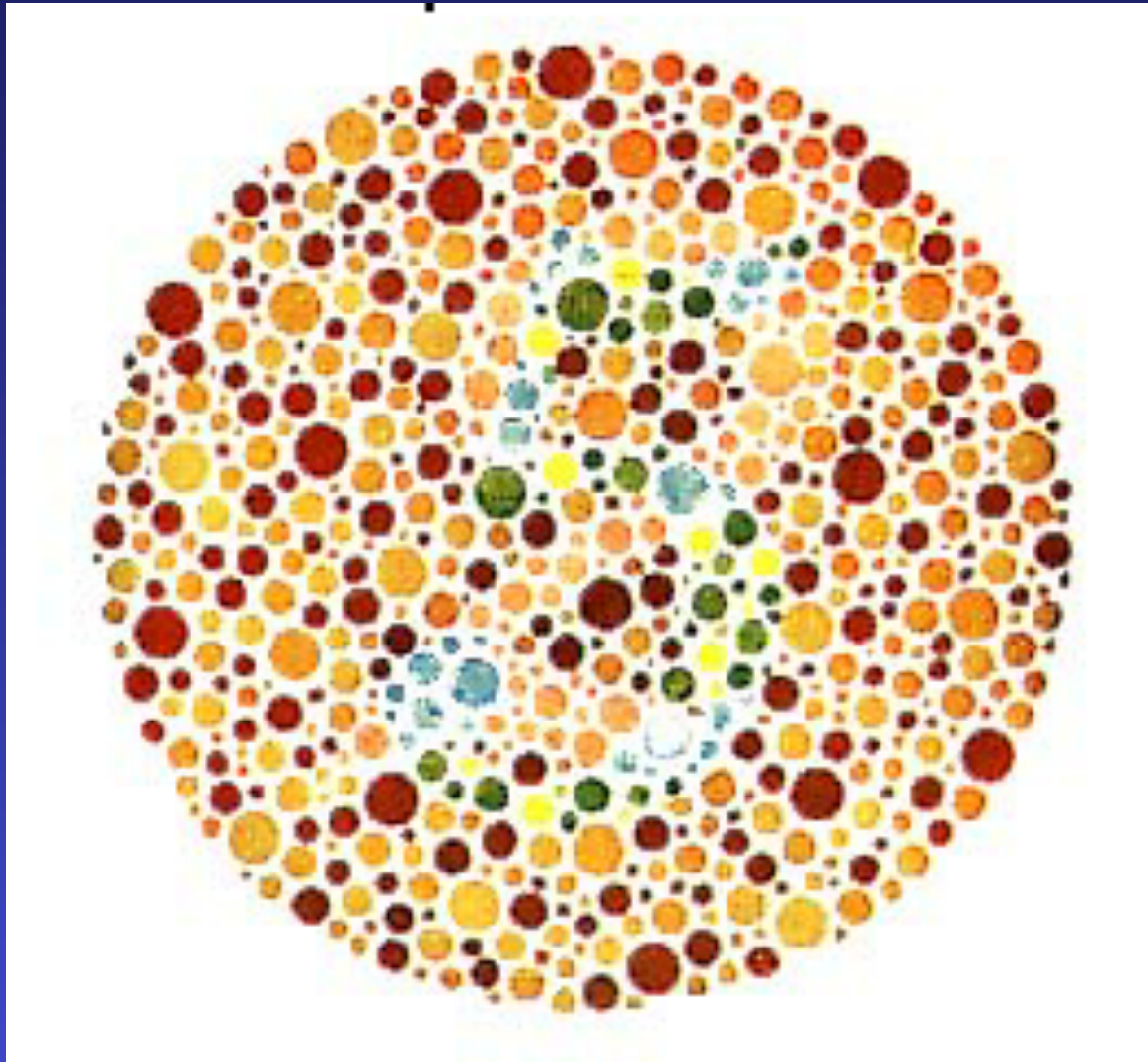
5. **Message**

- *What was the speaker's point?*

Value Your Slides

- ✓ Follow a single theme. Otherwise, too confusing.
- ✓ Each slide is a quantum.
 - *Make it count.*
 - *Don't speak too short or too long on a single slide.*
- ✓ Slides should be pretty, but not too pretty.
- ✓ Use large font sizes.
 - *If they can't see it, it is wasted!*
- ✓ ~10% of people are color-blind.
 - *Use white and yellow on dark background.*

Is it a 5 or a 2?



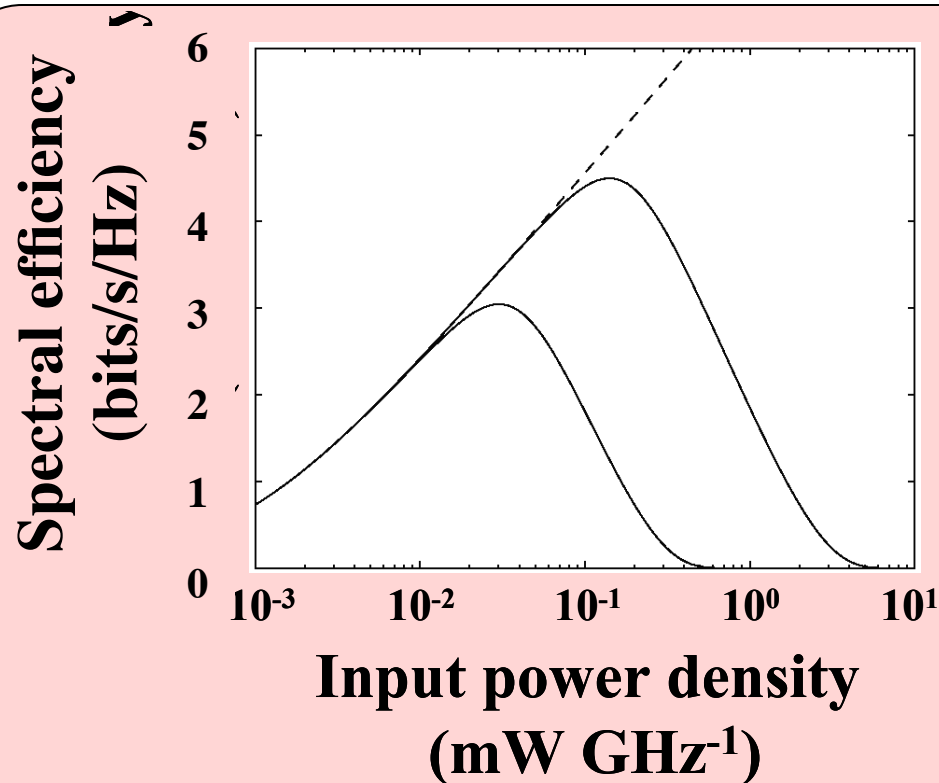
A Picture is Worth 1000 Words

- ✓ Avoid TEXT slides unless absolutely necessary.
- ✓ Figures that are appropriate for a journal are often NOT appropriate for a presentation.
 - *Avoid using legends. Label each curve separately.*
 - *A presentation figure can't be as complex as a journal figure.*
- ✓ *Any* idea can be expressed as a picture.
 - *See the following few example slides*

Spectral Efficiency

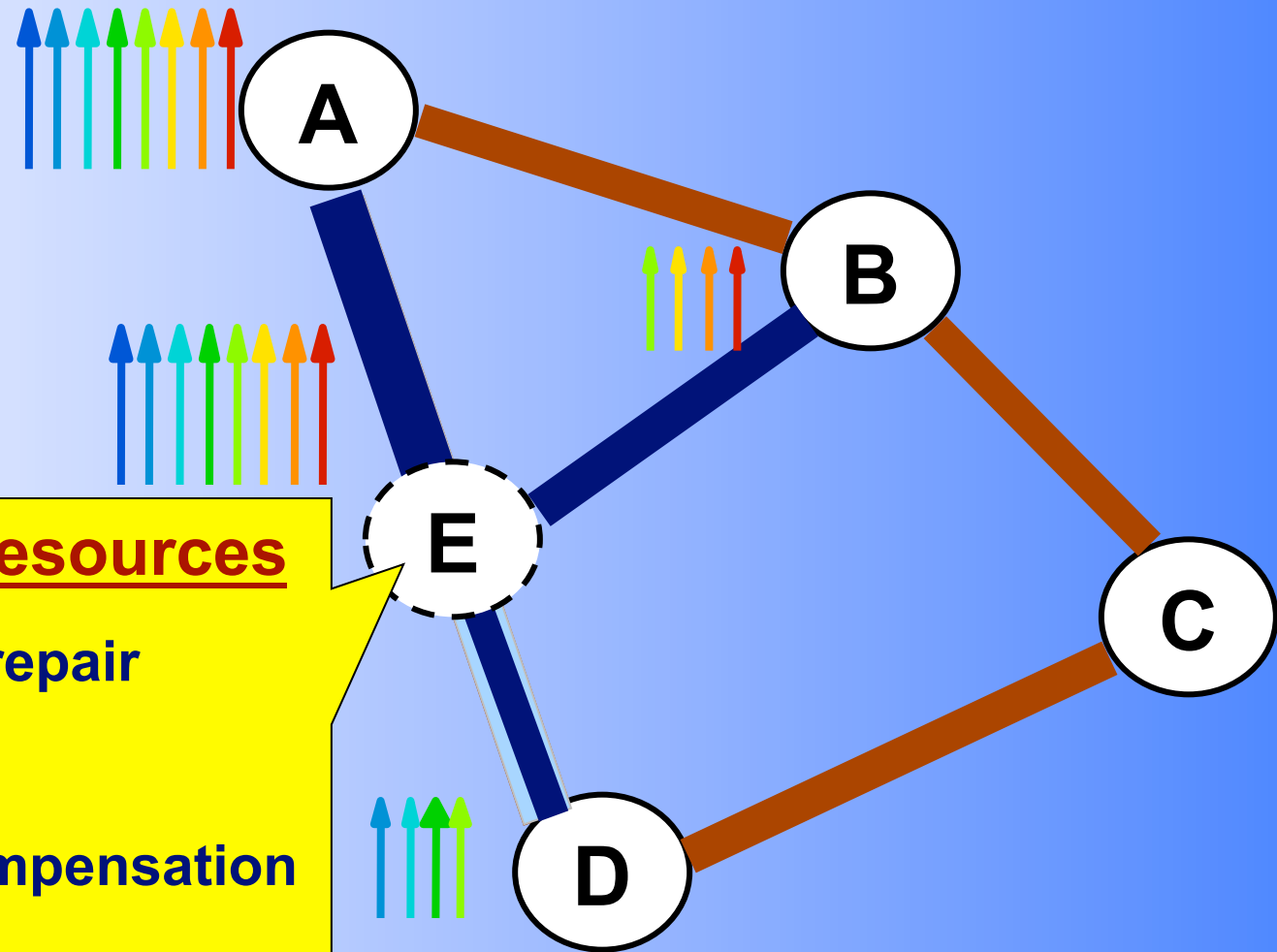
Spectral efficiency = Channel capacity per unit bandwidth

$$\text{Channel capacity} = (\text{Bandwidth}) \log(1 + S / N)$$



- Limited by optical amplifier noise and fiber nonlinearities.

Self-Assembled Networks



“Adaptive” Resources

- Diagnose and repair
- BW allocation
- Gain/Loss
- Dispersion Compensation
- λ -Routing
- Look-up tables

Automation + Intelligence + Monitoring

Keep the person out of the loop

List of Slides

- Title (usually advertised, so make interesting)
- Motivation (*Picture*)
 - Why should I listen to you for 10 minutes?
 - What would the field be like without your results?
 - Where's the pain?
- Outline
 - *List of Topics (usually 3-5)*
- Several Technical Slides
 - *Avoid text slides*
- Summary
 - *What are your 2-3 key points?*
 - *What do you want the listener to remember?*
 - *Remember: Don't rush the summary*

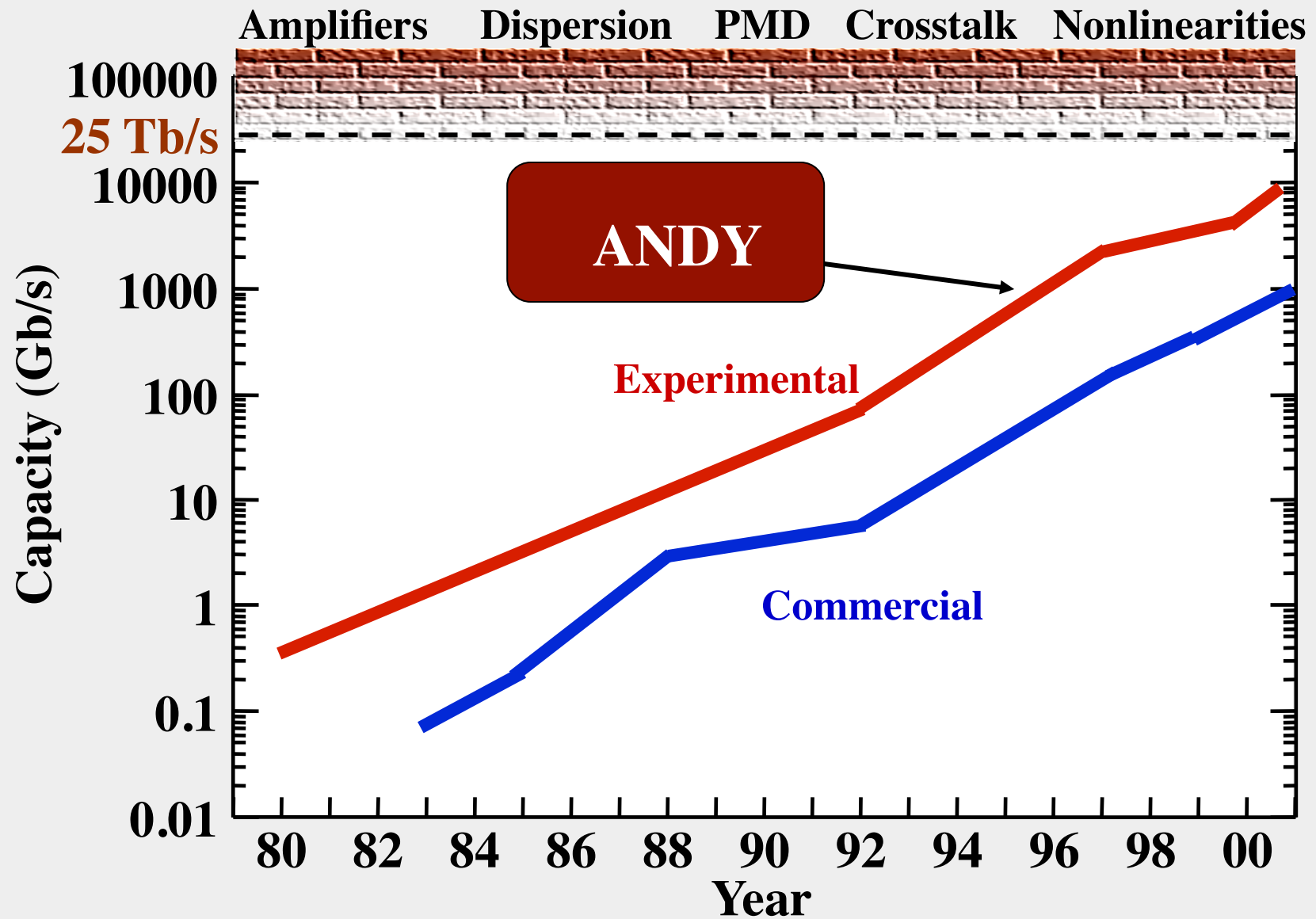
“Each Sub-topic Should Contain ...”

1. Introduce the importance of the topic.
2. Identify the key problem.
3. Give any pertinent background and prior art.
4. Describe your solution approach.
5. State main results (“We achieved 10X ...”).
6. Clearly articulate the impact, i.e., answer the question *“So what?”*

Some Helpful Hints (I)

- ✓ Share credit.
 - *Credit is an infinite reservoir.*
 - *“If someone who honors someone else, he himself is honored.” (Andy, Andy, Andy.)*
- ✓ Speak SLOWLY, loudly, and clearly.
- ✓ Look at the audience.
- ✓ Humor is good, but keep it appropriate to stature and length of presentation.
 - *Be culturally sensitive. (StarTrek)*
- ✓ In a long presentation, good to repeat **OUTLINE** multiple times.
- ✓ Transitions between VG' s are crucial.

Limitations of Optics



Some Helpful Hints (II)

- ✓ Appropriate dress:
 - A little above your audience.
- ✓ Don't distract your audience.
 - *Don't misspell anything.*
 - *Don't repeat odd phrases or motions.*
(“42”)
- ✓ Be familiar with talks that are before & after.

Some Helpful Hints (III)

- ✓ Animation can help or hurt the message.
- ✓ Worry about double entendres.
- ✓ You can learn as much from a bad talk as you can from a good talk.
- ✓ Be happy, comfortable (i.e., not pushy), and intellectually honest.
- ✓ Be serious AND light.
- ✓ Don't take yourself too seriously.

Formulae: “Be Afraid, Be Very Afraid!”

Can a listener understand it
in one minute?!?!?!?!?

$$E \propto c^2$$

Identify key trends only!!

Worry about offending 10% of your audience!!

- ✓ Use “might, may, could, ...”
- ✓ Avoid using “will, must, always, never, ...”
- ✓ Be careful about casual references to people or companies.
- ✓ Ask yourself, “What MIGHT someone think?”

The Epilogue (I)

- Never put anything on a slide on which you can't answer a question.
- Anticipate questions during practice.
- Always repeat the question from the audience: Everyone can hear, and the questioner can clarify.
 - ❖ Corollary: Never answer the wrong question.
- Keep answers to questions SHORT

The Epilogue (II)

- Thank audience for a question.
 - ✓ Give it credence, and never belittle the question.
- Never argue with a sniper questioner. You will lose.
 - ✓ *“Never argue with a pig. You both get dirty, but the pig likes it.”*
- Be a good audience member.
 - ✓ Ask a good “What if?” question.
 - ✓ Never embarrass the speaker!!!

The Three Most Important Words

“I DON’ T KNOW”

- You don’ t need to know everything.
- An audience can smell if you are “snowing” them.

Practice Makes “*Almost*” Perfect

- ✓ You will NEVER be a perfect speaker.
- ✓ You will spend your whole career trying to get better.
- ✓ You can learn from bad presenters as well as good presenters.
- ✓ Leave no audience member behind.
- ✓ This is just my opinion.
 - *Gather many opinions, and then learn what works for you.*

Summary

Learn good habits from many different people, keep trying to get better, and choose your own path to effective presenting skills.